

Detailed results for “A simheuristic for the cross-docking scheduling problem under uncertain truck arrival times”

1. Detailed comparison results of SimHGS against the reference algorithms

This section shows the detailed comparison results by the SimHGS and the reference algorithms, including TSim, SBDF, SBDB. Tables 1 to 3 show the comprehensive results on the three datasets, SetA, SetB, SetC, respectively. The “Inst.” column presents the name of each instance, where the number of inbound trucks, outbound truck and inbound doors and out bound doors can be inferred from their names. The columns *Best* and *AvgObj* indicate the best objective value and average objective value over 10 independent runs. *AvgT* shows the average running time in seconds. The best solutions among the compared results are highlighted in bold in the tables. In addition, the average value of best solution on an out-of-sample set of 20,000 scenario are given in column “E”.

Table 1 Detailed comparison result of SimHGS against TSim, SBDF and SBDB on SetA instances

Inst.	SimHGS				TSim				SBDF				SBDB			
	<i>Best</i>	<i>AvgObj</i>	<i>AvgT</i>	E	<i>Best</i>	<i>AvgObj</i>	<i>AvgT</i>	E	<i>Best</i>	<i>AvgObj</i>	<i>AvgT</i>	E	<i>Best</i>	<i>AvgObj</i>	<i>AvgT</i>	E
10x3A	5441.3	5441.3	8.8	5443.3	6181.1	6181.2	0.1	6150.9	5443.2	5450.7	21.4	5445.3	5443.2	5454.9	60.1	5445.3
10x4A	7533.6	7533.6	2.1	7528.4	7744.4	7744.4	6.1	7725.7	7533.6	7533.6	6.2	7528.4	7533.6	7533.6	11.1	7528.4
15x3A	10019.0	10019.0	14.5	10019.0	10027.9	10027.9	103.3	10027.9	10023.1	10023.5	4.4	10020.9	10023.1	10024.4	11.0	10020.9
15x4A	13226.2	13226.2	42.5	13227.1	13751.4	13751.4	14.3	13762.0	13238.3	13238.3	37.4	13238.1	13238.3	13238.3	130.2	13238.1
20x3A	17695.0	17695.0	128.8	17695.0	18191.5	18220.0	178.8	18191.4	17695.0	17695.0	88.3	17695.0	17695.0	17695.0	296.1	17695.0
20x4A	17805.8	17805.8	176.6	17794.8	18087.5	18092.7	614.7	18075.3	17805.8	17805.8	506.5	17794.8	17805.8	17806.1	519.9	17794.8
25x3A	70463.5	70464.3	968.3	70465.9	70666.8	70682.9	713.6	70662.4	70469.8	70472.3	464.1	70471.3	70469.8	70472.3	752.6	70471.3
25x5A	35221.3	35222.4	1192.3	35227.2	35920.2	36061.8	894.8	35925.5	35223.1	35227.2	927.1	35231.3	35224.8	35232.5	1404.8	35231.8
28x3A	65003.5	65003.5	304.0	65050.5	65435.9	65436.6	211.0	65433.4	65004.0	65013.7	736.8	65011.4	65004.0	65009.1	803.2	65011.4
28x5A	63293.8	63297.9	829.9	63297.0	64528.8	64716.7	1174.7	64533.1	63297.1	63311.6	1350.7	63301.5	63299.6	63332.7	1234.2	63303.6
30x6A	53201.5	53207.4	1145.5	53200.2	54360.9	54472.2	1048.7	54366.7	53217.4	53231.4	1539.9	53226.5	53231.3	53250.4	1518.5	53237.4
30x8A	61352.4	61359.5	1062.8	61362.3	62634.9	62919.4	1021.3	62670.2	61382.6	61638.8	1489.0	61390.9	61410.1	61664.6	1575.1	61428.1
35x6A	88253.4	88420.0	1475.5	88276.9	89484.8	89661.8	1232.0	89494.4	88406.9	88635.5	1608.5	88439.3	88648.5	88768.2	1499.5	88651.6
35x8A	83443.6	83487.8	1415.7	83429.3	84860.4	84914.8	1215.2	84844.7	83640.7	83728.7	1557.2	83622.6	83738.1	83832.8	1440.3	83713.3
40x6A	107222.3	107306.6	1623.5	107293.3	109285.7	109760.7	1641.6	109363.2	107351.6	107708.3	1493.4	107430.5	107696.0	107897.0	1289.3	107776.0
40x8A	118623.6	118892.1	1577.6	118657.0	120417.4	120889.4	1316.4	120436.7	119144.1	119248.4	1449.9	119161.4	119199.8	119430.0	1418.0	119203.0
50x6A	259656.7	259869.6	1653.7	259652.1	260943.6	261216.0	1657.8	260904.7	260061.1	260153.5	1296.9	260048.0	260151.2	260237.6	673.6	260146.8
50x8A	168321.2	168378.9	1522.3	168355.2	171459.9	171710.1	1577.1	171457.3	168610.9	168956.8	1617.9	168647.5	169022.6	169352.5	1092.5	169034.8
60x8A	256174.8	256684.3	1603.1	256202.0	260713.3	261217.1	1674.8	260810.0	258085.3	258652.0	540.4	258129.4	258085.3	258815.9	1165.0	258129.4
60x10A	258497.1	258632.3	1644.7	258447.2	263020.8	263635.6	1680.6	263001.0	260783.5	261486.0	1148.5	260765.7	260783.5	261695.9	1036.0	260765.7
80x8A	554099.4	554848.9	1300.5	554057.4	560289.4	561870.4	1216.8	560090.5	555576.9	555858.2	1258.3	555483.2	555576.9	556354.2	1403.3	555483.2
80x10A	373006.0	374263.0	1365.2	373042.9	383192.8	384059.2	1112.3	383124.1	376213.5	377380.9	1468.6	376286.0	377562.9	378674.3	1921.1	377586.2
100x10A	773276.7	774292.5	621.0	773151.4	783806.3	785425.3	895.4	783624.9	775503.2	777986.2	1946.0	775483.1	776606.7	780048.0	2469.8	776564.6
100x15A	749159.9	751312.6	1347.2	749120.0	762277.7	764966.5	1076.2	762309.2	755112.7	762027.5	2575.5	755144.3	765069.7	768359.6	3336.6	765039.5
#Wins	23	23			0	0			0	0			0	0		
#Ties	1	1			0	0			1	1			1	1		
#Losses	0	0			24	24			23	23			23	23		

Table 2 Detailed comparison result of SimHGS against TSim, SBDF and SBDB on SetB instances

Inst.	SimHGS				TSim				SBDF				SBDB			
	<i>Best</i>	<i>AvgObj</i>	<i>AvgT</i>	E	<i>Best</i>	<i>AvgObj</i>	<i>AvgT</i>	E	<i>Best</i>	<i>AvgObj</i>	<i>AvgT</i>	E	<i>Best</i>	<i>AvgObj</i>	<i>AvgT</i>	E
10x3B	17588.0	17588.0	24.1	17568.7	18470.4	18471.2	3.7	18432.5	17588.5	17588.5	38.7	17567.9	17588.5	17588.5	68.0	17567.9
10x4B	24588.2	24588.2	5.9	24549.1	25312.8	25312.8	0.9	25251.5	24592.1	24592.6	19.5	24554.8	24592.1	24592.5	48.5	24554.8
15x3B	37798.3	37798.9	935.0	37788.2	38652.6	38652.9	766.2	38661.5	37801.0	37810.4	935.7	37788.8	37799.6	37812.0	808.9	37788.9
15x4B	50542.0	50549.1	1161.6	50566.1	51587.3	51587.3	214.8	51619.6	50561.1	50589.5	955.5	50586.4	50561.1	50600.1	1199.5	50586.4
20x3B	62467.7	62471.0	1096.7	62481.1	64257.9	64278.4	1159.7	64277.5	62492.8	62580.5	1173.5	62494.5	62500.7	62580.4	1225.0	62502.8
20x4B	67575.6	67688.6	919.7	67545.8	68588.8	68619.5	663.9	68559.6	67644.1	67743.7	1168.5	67604.5	67705.0	67803.8	1313.8	67666.1
25x3B	241551.0	241605.6	1094.7	241567.5	242378.5	242460.4	1083.9	242363.6	241556.0	241628.7	1427.5	241570.1	241578.8	241675.2	1404.7	241592.1
25x5B	114512.0	114671.1	1505.2	114496.9	116520.1	116943.0	1157.7	116547.9	114696.2	115009.7	1495.3	114694.1	114884.7	115286.2	1588.6	114875.6
28x3B	241918.3	241992.3	1234.6	241922.1	243397.4	243497.8	929.2	243407.9	242009.2	242101.6	1437.8	242019.8	242076.2	242189.0	1436.0	242079.8
28x5B	195827.5	196148.6	1159.6	195835.5	198071.8	198211.7	1174.2	198065.7	196226.1	196464.3	1627.3	196259.7	196378.8	196556.9	1560.9	196401.9
30x6B	194666.0	194897.3	1560.6	194757.9	196919.7	197438.6	1292.7	197075.4	195332.3	195612.5	1526.3	195457.0	195332.3	195756.7	1448.9	195457.0
30x8B	201236.2	201480.0	1418.5	201248.1	203131.0	203481.5	1383.3	203169.7	201388.8	202052.5	1402.2	201382.6	201825.1	202415.0	1259.1	201837.1
35x6B	305978.9	306277.4	1495.4	305946.9	308568.0	309012.5	1450.4	308539.3	306921.5	307200.5	1693.2	306883.1	307007.9	307398.8	1420.8	306966.9
35x8B	268297.2	268486.7	1498.9	268304.2	271592.2	272157.5	1467.6	271634.0	268679.7	269094.8	1401.5	268652.3	269187.1	269704.2	1552.0	269166.2
40x6B	347541.0	347726.1	1551.0	347537.3	350919.6	351400.3	1568.5	350929.3	348585.6	349070.6	1351.6	348577.1	349022.7	349565.6	1383.8	349018.5
40x8B	362918.4	363447.9	1524.9	362944.5	367599.4	368369.7	1477.6	367731.1	364652.3	365122.9	1348.2	364685.7	365264.9	365723.0	1234.7	365307.5
50x6B	831909.3	832832.3	1634.2	831868.7	837326.8	837701.5	1432.7	837337.5	834484.9	834921.4	960.4	834526.2	834781.3	835140.6	1083.3	834841.1
50x8B	525307.9	526001.3	1458.2	525345.2	530497.0	531500.2	1509.9	530572.0	526954.1	528372.5	1482.4	526969.4	528601.6	529398.5	1457.7	528718.1
60x8B	941062.9	942626.9	1549.9	941083.0	947638.0	949489.9	1252.8	947501.1	945186.4	946507.6	1698.8	945161.8	948450.8	950365.5	2112.1	948426.4
60x10B	854676.4	855992.5	1270.8	854579.7	863732.0	865412.8	1261.7	863712.9	859524.4	861357.2	1643.2	859605.8	862870.8	868942.6	2076.9	862824.6
80x8B	1821242.1	1821761.9	1353.0	1821201.3	1830213.9	1833855.3	1210.3	1830054.3	1832637.2	1835033.1	2701.6	1832562.4	1832637.2	1835033.1	4131.1	1832562.4
80x10B	1337100.2	1337628.9	1037.6	1336938.5	1350149.6	1350599.2	1050.5	1350184.9	1357838.9	1360622.8	2731.4	1357784.3	1357838.9	1360622.8	5116.7	1357784.3
100x10B	2620042.1	2622073.9	1191.5	2620083.7	2635733.5	2638723.0	844.8	2635621.6	2648080.6	2652663.8	9165.1	2647936.3	2648080.6	2652663.8	18134.4	2647936.3
100x15B	2525547.2	2531717.2	1294.6	2525141.9	2542112.9	2545782.7	1304.1	2542001.0	2602788.3	2618073.1	7228.6	2602501.2	2602788.3	2618089.7	17918.8	2602501.2
#Wins	24	24			0	0			0	0			0	0		
#Ties	0	0			0	0			0	0			0	0		
#Losses	0	0			24	24			24	24			24	24		

Table 3 Detailed comparison result of SimHGS against TSim, SBDF and SBDB on SetC instances

Inst.	SimHGS				TSim				SBDF				SBDB			
	<i>Best</i>	<i>AvgObj</i>	<i>AvgT</i>	E	<i>Best</i>	<i>AvgObj</i>	<i>AvgT</i>	E	<i>Best</i>	<i>AvgObj</i>	<i>AvgT</i>	E	<i>Best</i>	<i>AvgObj</i>	<i>AvgT</i>	E
10x3C	43.8	47.1	960.0	51.2	441.8	516.2	1009.4	438.9	44.7	50.8	621.4	46.0	44.7	50.8	614.3	46.0
10x4C	17.0	19.7	1023.0	23.2	167.4	225.6	830.1	175.3	17.0	25.3	641.5	23.2	16.0	23.9	949.3	20.1
15x3C	8264.0	8342.9	297.2	8288.9	8612.6	8635.5	486.6	8636.1	8328.8	8493.8	416.9	8284.3	8278.9	8461.5	491.1	8287.4
15x4C	374.6	604.8	907.7	353.4	1626.6	1763.2	790.2	1607.0	529.9	714.9	924.9	508.7	580.6	749.9	921.6	560.1
20x3C	29764.9	30018.9	784.1	29791.1	29627.7	30074.7	894.4	29690.9	29626.5	29980.6	1066.7	29695.5	29625.2	29957.5	1112.6	29702.0
20x4C	5248.0	5707.4	305.2	5193.2	5724.4	5940.7	1204.4	5685.2	5363.9	5826.3	279.4	5346.3	5363.9	5826.3	326.7	5346.3
25x3C	82328.8	82564.9	1060.1	82390.3	82469.8	82834.1	916.4	82567.7	82114.8	82734.6	887.6	82179.4	82114.8	82771.6	866.4	82179.4
25x5C	43721.6	44134.4	1488.8	43829.2	45434.5	45700.1	1065.5	45533.9	43713.2	44182.3	1202.8	43800.0	43850.8	44405.4	1578.2	43906.6
28x3C	121351.1	122520.5	1198.0	121631.9	120168.9	122306.9	853.0	120521.3	121554.9	122967.1	1066.9	121856.2	121554.9	122949.7	1179.9	121856.2
28x5C	42244.0	43089.0	1425.1	42307.7	43433.0	43977.0	1267.4	43476.7	42963.6	43675.2	1404.4	42947.2	43610.6	44301.3	1379.8	43692.1
30x6C	36350.6	36685.2	1488.1	36448.9	38893.5	39326.2	1445.2	38973.4	37066.9	37838.2	1480.6	37098.9	37399.8	38222.9	1554.6	37571.1
30x8C	35803.9	36256.5	1511.2	35853.2	37900.3	38810.1	1192.5	37937.8	36327.5	37040.4	1373.6	36318.9	36947.7	37510.8	1356.2	37037.0
35x6C	119484.0	120939.0	1597.9	119510.2	123462.7	125167.7	1285.8	123600.0	122853.9	124383.4	1224.5	122900.4	124407.4	125496.9	1166.3	124466.8
35x8C	114372.7	115674.5	1426.0	114316.4	116254.5	118367.9	1242.5	116150.6	117696.4	118764.7	1097.8	117863.3	118116.8	119264.6	999.4	118314.6
40x6C	181321.3	182234.3	1543.9	181231.9	183514.7	184858.1	1365.4	183444.1	184327.0	186213.6	1143.0	184212.1	186034.7	186976.6	959.7	186061.6
40x8C	180282.8	182311.0	1038.2	180337.5	183297.7	185616.8	1225.6	183521.0	183813.5	185168.0	1185.8	183772.0	185336.2	186454.6	1439.3	185291.4
50x6C	425567.2	427706.7	1183.5	425714.7	430280.0	432936.9	1021.8	430400.1	427878.7	434553.3	1311.9	427920.1	433805.1	439072.2	1463.1	434264.5
50x8C	417126.0	418412.3	1205.1	417445.2	419300.4	423440.7	1067.7	419659.2	422197.3	425011.9	1288.4	422363.7	426232.0	428225.1	1318.3	426270.1
60x8C	531970.1	534373.0	1252.2	531943.3	536376.6	538181.8	684.1	536289.6	538146.4	539889.1	1352.5	538061.6	540524.4	550848.0	1711.9	540501.1
60x10C	532976.9	535436.8	1136.6	533136.6	539582.8	540844.1	1178.3	539615.9	538982.3	542276.1	1192.9	538753.2	539400.9	552551.0	1770.9	539120.1
80x8C	1096037.7	1098380.3	1121.5	1095927.7	1104855.6	1107698.1	1025.1	1104757.4	1121563.6	1142961.2	2288.6	1121165.2	1171623.4	1210631.8	3492.3	1171464.0
80x10C	1098289.2	1100530.5	1185.1	1098116.4	1105294.6	1108670.9	972.1	1105334.4	1126571.3	1161996.6	2541.1	1126461.6	1172733.8	1200447.9	3887.5	1172570.9
100x10C	2230299.8	2235967.0	857.0	2229970.4	2243621.8	2245878.0	721.7	2243292.8	2340771.0	2397956.2	5126.9	2340610.6	2340771.0	2397956.2	8312.8	2340610.6
100x15C	2239781.4	2244887.0	868.4	2239764.5	2249184.4	2252212.5	1305.2	2248922.3	2407298.8	2442178.6	6456.4	2406960.4	2407298.8	2442178.6	9953.3	2406960.4
#Wins	19	22			1	1			2	2			3	1		
#Ties	0	0			0	0			0	0			0	0		
#Losses	5	2			23	23			22	24			21	23		